Task 03: Networking Basics for Cyber Security

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Objective

The goal of this task is to understand basic networking concepts and analyze live network traffic using Wireshark. This helps in learning how data flows over a network and how attackers can observe insecure traffic.

Tools Used

- Wireshark
- Web Browser (for generating traffic)

Networking Concepts Learned

- **IP Address**: Unique address of a device on a network
- **MAC Address**: Physical address of a network interface
- **DNS**: Converts domain names into IP addresses
- **TCP**: Reliable, connection-oriented protocol
- **UDP**: Fast, connectionless protocol

Packet Capture Observations

1. **TCP Three-Way Handshake**
   Observed SYN, SYN-ACK, and ACK flags which establish a TCP connection between client and server.
2. **DNS Queries**
   DNS packets were captured showing domain name resolution when accessing websites.
3. **Plain-text vs Encrypted Traffic**
   - HTTP traffic may expose readable information.
   - HTTPS traffic is encrypted and cannot be read directly, making it more secure.

Conclusion

This task provided hands-on experience in capturing and analyzing network traffic. It demonstrated how insecure protocols can expose sensitive data and why encrypted communication is critical for cyber security.